

DEFINITIONS

Incident: A situation with the potential to cause serious harm to a person. Generally, the outcome results in first aid treatment.

Accident: A situation that results in medical attention (treatment beyond first aid) and/or lost time (missed work).

Supervisor: A supervisor may be a dean, department head, director, manager, administrator or any other faculty or staff person who is in charge of one or more employees.

RESPONSIBILITIES

Employee: Immediately report all work-related incidents and accidents to your supervisor and participate in the investigation process, as needed.

Supervisor: Report incidents and accidents as directed by the OSU Insurance and Risk Management Services [Worker's Compensation Resources](#) website. In the event of lost time, an [Accident Investigation Report](#) shall be initiated by the employee's immediate supervisor within 48 hours of the accident. If the immediate supervisor is absent, another manager from the same operational unit may provide assistance. It is incumbent upon the employee's immediate supervisor to ensure that the corrective measures are completed, documented, and communicated to all affected employees, and incorporated into the appropriate policy, procedure, or safety program in order to prevent future occurrences.

ACCIDENT INVESTIGATIONS

An accident investigation systematically identifies event details and causal factors to determine corrective measures.

As only 2% of all workplace incidents are thought to be unpreventable*, the primary purpose of an investigation is to prevent future occurrences, not to place blame.

*OR-OSHA 1207-05

Beyond the primary purpose, the information obtained through the investigation should be used to update and revise the investigator's inventory of hazards, and/or the relevant safety program(s) for hazard prevention and control. For example, a [Job Hazard Analysis](#) may be generated or revised and employees (re)trained to the extent that it fully reflects the recommendations made in the investigation report. Further, implications from the root cause(s) of the accident should be analyzed for their potential impact on other operations and procedures.

ACCIDENT / INCIDENT CAUSAL FACTORS

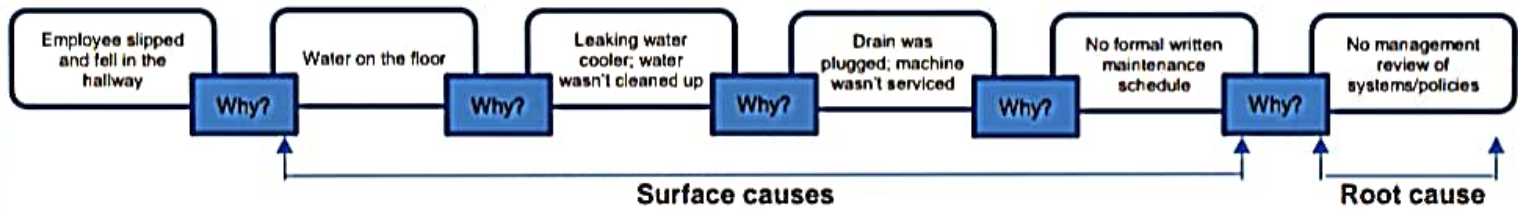
There are two major components that contribute to the cause of an accident / incident; surface cause and the root cause.

The **surface cause** is the condition or act that directly caused the incident. An example of a surface cause is a small spill of oil on the floor that someone slipped on.

The **root cause** is the system failure that allowed the surface cause to occur. For example, a root cause may be a lack of preventive maintenance that resulted in the fork truck leaking oil on the floor. A thorough investigation will reveal the root cause of the incident. Corrective measures that address the root cause have the greatest potential to prevent accident / incident recurrence.

BASIC STEPS FOR CONDUCTING AN INVESTIGATION

- **Secure the scene.**
 - This is the beginning of your analysis. Your primary goal is to secure the scene as soon as possible in order to prevent further injuries, ensure the well-being of the affected employee, and to protect any critical physical clues from being spoiled.
- **Collect the facts.**
 - Focus on finding the facts about the event. Remember to gather valid information without drawing conclusions or assigning blame. Document your observations. Take photos and check video surveillance if available. Interview employees and witnesses. Review relevant records, such as maintenance, training, policies, procedures, etc.
- **Develop the sequence of events.**
 - Review and accurately arrange the gathered information to determine the order of events. Constructing an accurate timeline may be critical to an effective analysis. Document what happened before, during, and after the event. Arrange this information to accurately determine the order of events.
- **Determine potential causal factors.**
 - Every accident/incident is caused by a set of contributing factors. These factors represent the surface or root causes that led to the event. The goal is to identify these by analyzing how or why each consecutive event happened. Use the following diagram as an example.



Surface causes: unsafe behaviors and hazardous conditions; Root cause: underlying problems with policies and/or procedures.

- **Recommend or implement corrective measures.**
 - Your recommendations should be relevant and concise.
 - Identify, either independently or as part of a collaborative effort, and describe:
 - The recommended corrective measure(s),
 - Who will be responsible for implementation, and
 - The anticipated completion date.
 - Follow up to ensure that:
 - The corrective measures are implemented within the appropriate time frame, and
 - Incorporated into the appropriate policy, procedure, or safety program in order to prevent future occurrences.
- **Communicate the outcomes of the investigation.**
 - Train your employees on what changes will be / have been implemented.
 - Share your experience with peers so they, too, may enjoy a safer and more productive work environment.

Reporting

A written report should be generated that contains the following sections:

- ***Statement of the problem.*** This section should include:
 - o a review of the incident
 - o a summary of injuries, lost time, and equipment and/or property damage.
- ***Review of the data.*** This section should include:
 - o a summary of witness statements
 - o a summary of relevant findings concerning the accident and work history of affected employees and the operation of machinery or equipment
 - o a storyboard with photographs or sketches
 - o an overview of existing, written company policies or directives.
- ***Causes.*** This section should list the direct, indirect, and contributing causes that have been affirmed by the data. A reference should be made to the data that support each cause.
- ***Recommendations.*** These should be based directly on each of the noted causes. These recommendations could include the following:
 - o more or improved training for employees
 - o new company policies or directives, or better clarification or dissemination of existing ones
 - o improved communication between employees, supervisors, and management
 - o design or operation changes or improvements to machines, equipment, or processes
 - o different or improved safety equipment
 - o different or improved protection from natural phenomena or disasters
 - o different or improved systems to account for possible physical, physiological, or psychological limitations of employees, customers, or others.

1.16 Hazard Identification During Operation (HAZOP)

Hazards arise due to deviations from normal process. There always exist deviations from the design intent. This is applicable to the existing and new process plants. The main purpose of HAZOP study is to identify the potential hazards and the relative operability problems that arise due to the perceived deviations. HAZOP analysis identifies all possible hazards, operational problems, recommends changes, and identifies areas that require further detailed studies. For conducting a HAZOP analysis, up-to-date Process Flow Diagram (PFDs) is required. It also requires Process and Instrumentation Diagram (P&IDs), detailed equipment specifications, details of materials and mass and energy balances. A team of experts who are experienced in a similar plant, along with the technical and safety professionals conduct HAZOP study.

1.16.1 HAZOP Objectives

HAZOP studies are carried out to identify the following:

- Any perceived deviations from intended design/operation
- Causes for those deviations
- Consequences of those perceived deviations

- Safeguards to prevent the causes and mitigate consequences of the perceived deviations
- Recommend actions in the design and operation to improve safety and operability of the plant

1.16.2 Common Application Areas of HAZOP

HAZOP is primarily used in chemical industries to estimate hazards that arise during operations; one such example can be seen in hazard studies carried out in Flixborough disaster, 1974. It is a chemical plant in the United Kingdom, which manufactures caprolactam that is required to manufacture nylon. This incident occurred due to the rupture of a temporary by-pass pipeline carrying cyclohexane at 150°C, which leaked and it set into a fire. Within few minutes, after the initiation of fire, about 20% of the plant's inventory got burnt and resulted in the spread of a vapor cloud over a diameter of about 200m. This resulted further in an explosion of a hydrogen production plant located nearby, which showed a cascading effect of the consequences. Another similar example where HAZOP studies were applied successfully was to study the consequences of explosion at a Rocket-fuel plant located at Nevada, Las Vegas, United States, as shown in Figure 1.8. The plant was destroyed in few seconds after the initiation of explosion. The wind storm

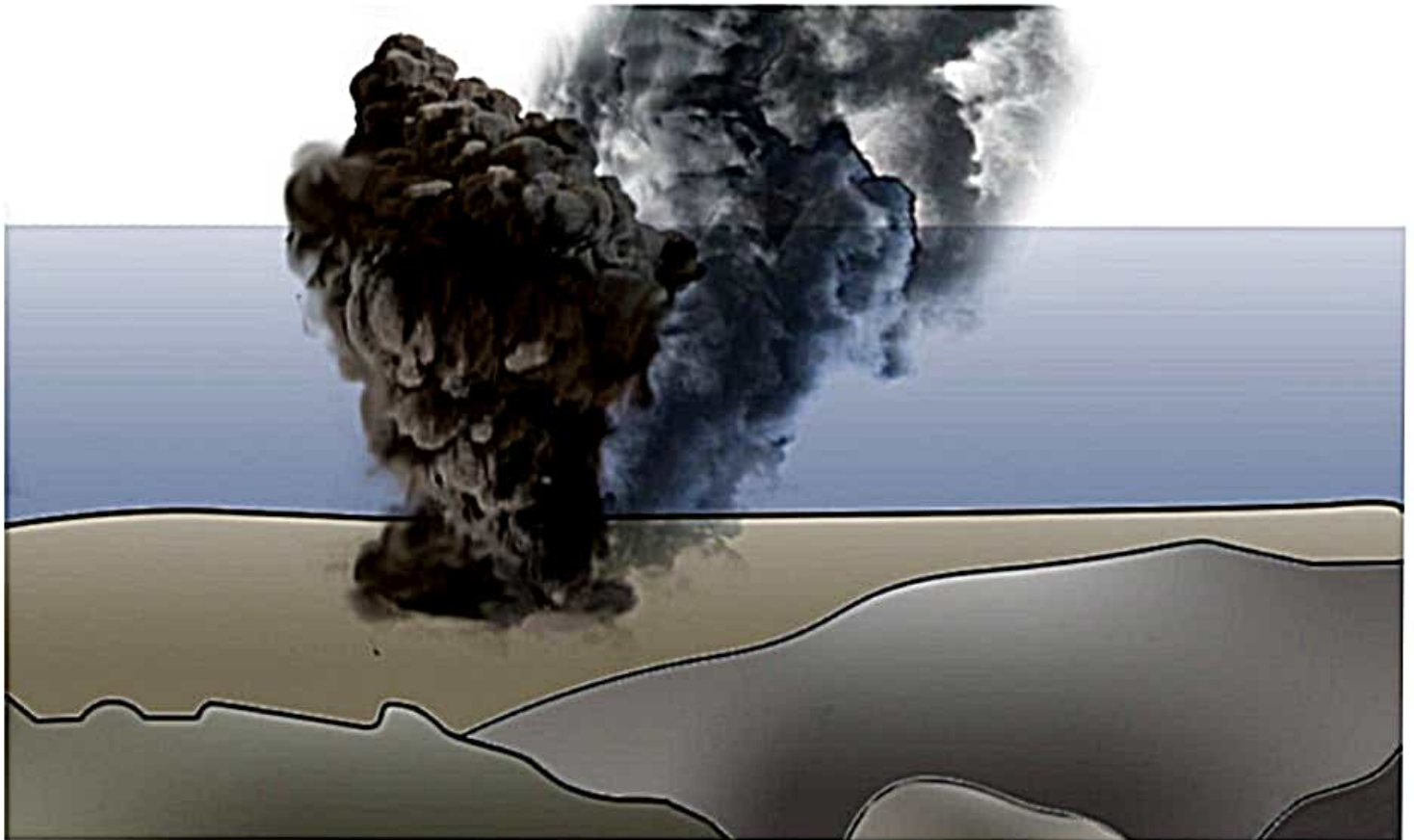


Figure 1.8 Explosion at Rocket-fuel plant located at Nevada, Las Vegas

1.16.3 Advantages of HAZOP

HAZOP supplements the design ideas with imaginative anticipation of deviations. These may be due to equipment malfunction or operational error. In the design of new plants, designers sometimes overlook few issues related to safety in the beginning. This may result in few errors. HAZOP highlights these errors. HAZOP is an opportunity to correct these errors before such changes become too expensive or impossible. HAZOP methodology is widely used to aid loss prevention. HAZOP is a preferred tool of risk evaluation because of few reasons: (i) easy to learn; (ii) can be easily adapted to almost all operations in the process industries; (iii) is a common method in contamination problems rather than chemical exposure or explosions; and (iv) requires no special level of academic qualification to perform HAZOP studies.

HAZOP studies examine the full description of the process thoroughly. It systematically questions every part of it to establish the perceived deviations from that of the design intent. Once identified, an assessment is made to estimate the consequences of such deviations. If considered necessary, action is taken to rectify the situation in the beginning itself. Though the method is imaginative, but it still is systematic. It is more than a checklist type of review. It encourages the team to identify possible deviations and helps to trace all of them under the operational conditions. HAZOP penetrates into greater depth of risk analysis of any process plant. HAZOP, applied on the same type of plant, repeatedly improves safety, which is quite important. Potential failures that were not noticed in the earlier studies can be easily highlighted using HAZOP.

1.17 Steps in HAZOP

Step 1: Define the design intent

Defining the design intent is the first step in a HAZOP study. Let us explain the design intent using some examples. Consider the following:

1. Suppose there is a plant in operation, which has to produce certain tons of chemical per year.

2. an automobile unit has to manufacture certain number of cars every year.
3. a plant has to process and dispose certain volume of effluent per year.
4. an offshore plant has to produce certain barrel of oil every year.

In all the cases, equipments are designed and commissioned to achieve the desired production capacity. In order to do so, each item like the equipments, pump, length of pipe work will need to be consistently functional in a particular (desired) manner. This is the design intent for that particular item and not the machinery or production capacity.

Step 2: Identify the deviations

For understanding the deviation in design intent, let us consider another example. A plant requires continuous circulation of cooling water at temperature $x^{\circ}\text{C}$ and at xxx L/h . Cooling of the process is done by heat exchanger. For effective functioning of the plant, effective working of heat exchanger is mandatory. The design intent is the effective working of the heat exchanger. If the water supplied for circulation becomes greater than $x^{\circ}\text{C}$, this would affect the production and hence this is the deviation. Note the difference between the deviation and its cause. For example, failure of pump would be a cause and not a deviation.

1.18 Backbone of HAZOP

The backbone of HAZOP studies is the keywords that are used in the study. There are two types of keywords: primary and secondary. **Primary** keyword focuses the attention on a particular aspect of the design intent or an associated process condition. **Secondary** keywords suggest possible (perceived) deviations from that of the design intent; when combined with that of the primary keywords, they intent the required meaning. As HAZOP revolves around the effective use of these keywords, it is necessary to understand their meaning and usage.

Primary keywords reflect both the process design intent and operational aspects of the plant. Examples are : FLOW, TEMPERATURE, PRESSURE, LEVEL, SEPARATE, COMPOSITION, REACT, MIX, REDUCE, ABSORB, CORRODE, ERODE, etc. These keywords sometimes may be confusing. For example, let us take the word CORRODE. One may assume that the intention is that corrosion should occur as it refers to the design intent. Most of the plants are designed with the design intent that corrosion should not occur during the life span; or if it is expected, it should not exceed a certain rate. An increased corrosion rate would result in the deviation from the design

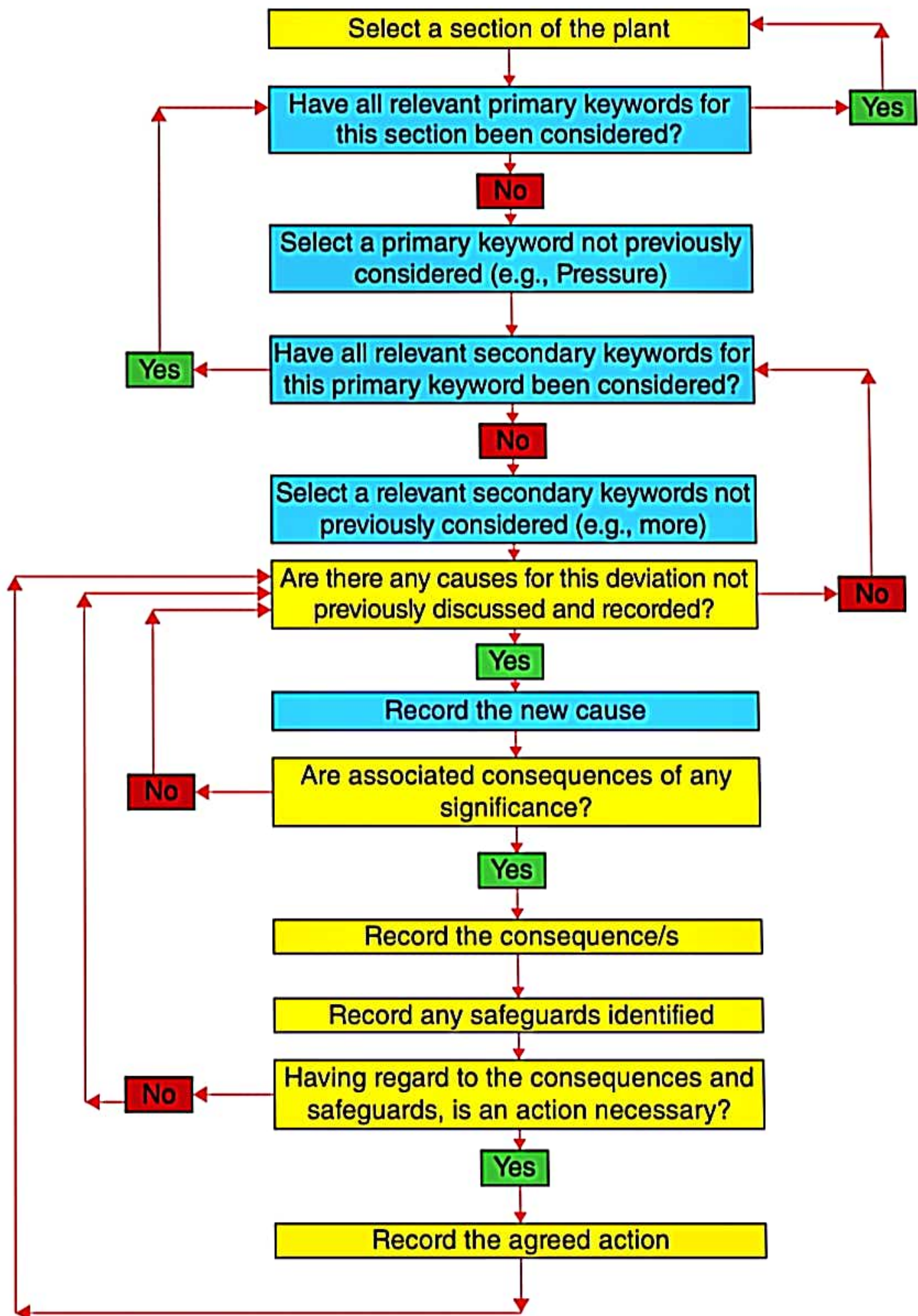


Figure 1.10 HAZOP flowchart

1.22 When to Do HAZOP?

HAZOP studies are generally carried out to identify potential hazards and operability problems caused by deviations that arise from the design intent. In particular, if there are major deviations made during any recent modifications made in the process line, then the changes should be verified for their safety through HAZOP studies. As a general practice in oil and gas industries, HAZOP studies are carried out at periodic intervals of not later than 6 months. HAZOP studies should preferably be carried out as early in the design phase as possible because this influences the changes in the design if deemed fit. But unfortunately, a good HAZOP study can be carried out only on the availability of a complete design. As a compromise, HAZOP is usually carried out as a final check when the detailed design is completed. HAZOP studies may also be conducted on an existing facility to identify the modifications that should be implemented to reduce risk and operability problems. Following situations generally necessitates HAZOP studies:

- At the initial concept stage when the design and detailed drawings are available.
- When the final P&ID are available.
- During the construction and installation to ensure that valid recommendations are implemented.
- During commissioning of the plant.
- During operation of the plant to ensure that the plant emergency and operating procedures are regularly reviewed and updated as required by OSID norms.

1.22.1 Types of HAZOP

Different types of HAZOP studies are conducted depending upon the objective of the said problem. HAZOP reports should follow a set of standard procedures to make it valid under legal challenges (IEC 61882; Crawley et al., 2000; Kyraikdis, 2003). The following list explains the types along with their applicability.

Process HAZOP: A technique that was originally developed to assess plants and process systems. This is quite a common type that is being practiced in oil and gas industries.

Human HAZOP: A “family” of specialized HAZOPs. More focused on human errors than technical failures. Usually conducted only on violations of work permits or report of a bulk of near-miss events.

Procedure HAZOP: Review of procedures or operational sequences, sometimes denoted Safe Operation Study (SAFOP). This is usually carried out while a major deviation in the process line is proposed.

Software HAZOP: To identify possible errors in the development of software. This is useful to analyze the hazards that may arise from the failure of automated control systems. This is essential for all electric and electronic control systems and is often practiced in oil and gas industries.

4.19 Process Hazard Analysis (PHA)

Every employer must perform an initial PHA on the covered processes. PHA studies should account for various complexities of the process and must identify, evaluate, and control the hazards involved in the process. The employer must use one or more of the following methodologies:

- What-if
- Checklist
- What-if/checklist
- Hazard and Operability Study (HAZOP)
- Failure mode and effects analysis
- Fault tree analysis
- An appropriate equivalent methodology

The PHA must address hazards present in the process while identifying any previous incident, which had a likely potential for catastrophic consequences in the workplace. Engineering and administrative controls are applicable to the hazards and their interrelationships such as appropriate application of detection methodologies to provide early warning of releases. A qualitative assessment of facility siting, human factors, and a range of the consequences of the failure of engineering and administrative controls need to be an inherent part of PHA.

Process hazard analysis must be performed by a team with expertise in engineering and process operations. The team must include at least one employee who has experience and knowledge of the process being evaluated. In addition, one of the team members must be knowledgeable in the specific PHA methodology being used. The employer must establish a system to promptly address the team's findings and recommendations; assure that the recommendations are resolved in a timely manner and that

the resolution is documented. Once a PHA is carried out and the report is prepared, it becomes a legal binding on the employer to implement the recommendations, at least in a phased manner. It is also mandatory for the employer to document the action taken on each recommendation and submit to safety audit every year for approval. At least, once in every 5 years, PHA must be updated and revalidated by a team to assure that it is consistent with the current process. Employers must retain process hazard analyses and updates or revalidations for each covered process for the life of the process.

Process Hazard Analysis (PHA)

Process Hazard Analysis (PHA) : Through process hazard analysis, we identify and evaluate the hazards of processes involving high-risk chemicals and establish a systematic approach to its control, which plays an important role in preventing chemical accidents.

The main objective of process hazard analysis is to identify and evaluate incidents and then consider their control, and to reduce the likelihood of consequences of accidents caused by hazardous substances, the main objective of PHA.

Therefore, in a place where the level of risk is high during the work, process hazard analysis is considered at such a place, which proves to be a good option to do the work systematically and prevent accidents.

Process Hazard Analysis Concept

As far as process hazard analysis is concerned, it plays an important role in improving safety within any organization and in preventing the effects of unplanned release of chemicals and prevention and decision making. So when thinking about PHAs, the first simple question to ask is “Is risk at the workplace acceptable?”

The PHA should always be performed by a well-trained, experienced and competent team and should include a process engineer. This is done so that there will be no output commitments for the tasks and people will not think that it is a waste of time.

If the team member lacks knowledge and experience with him then the result of PHA can be useless and very bad, as it is not implemented properly at workplace.

When a person is not confident while designing a PHA that the process and technology of the PHA designed is not efficient enough, they are unable to implement an effective process hazard analysis.

All the organizations where the employees are working, then the expenses on accidents that happen at such places are high and most of the organizations understand that definitely it is very expensive and it is unnecessary cost which should be prevented. could.

Therefore, it is very important to understand the exact cause of accidents, to find appropriate solutions to prevent them and prevent recurrence of damages.

As far as the PHA is concerned, it is basically a model that plays an important role in determining the adequate and effective control of the identified potential hazards and risks within the organization. That is why it is very important to understand the exact purpose of whatever program is conducted within the organization to prevent harm.

Therefore, you have to understand that it is not enough to be safe by investing sufficient amount to reduce the risks, but by effectively implementing the PHA, the risks can be reduced and the loss of property can be prevented.

Principles of Process Hazard Analysis

Process hazard analysis is done to analyze and identify threats within any organization and also to analyze the importance of threats.

Through this, the security measures within the organization are seen whether it is adequately given according to the task or not. And whether a systematic and comprehensive review of the process has been done to reduce them when necessary find out through it.

PHA is not meant to review the design and quality of any work or the operation of the work. This process should take place before PHA meetings to ensure that everyone has accepted the functionality of the PHA design.

While working within the organization we see or experience that most of the accidents are due to equipment failure or improper operation of the equipment. And some accidents are caused due to failure of risk control.

There are some such errors within the organization which indicate accidents such as – maintenance error, alarm failures, shut down system failures, events occur due to source of ignition or failure of management system.

Therefore, in order to avoid such errors, we design an engineer system which provides freedom from all the above mentioned dangers.

When do You Perform PHA

- When a new project is started.
- After making changes in the processes that are going on at the workplace.
- To assess process risks and changes in the operation of facilities every year, two years, or five years of existing facilities.

In terms of a proper PHA assessment, it focuses on collectively reviewing the risks of design and safety as well as operational teams and incorporating input approaches to prevent accidents and develop procedures.

We see that most of the organization prefers techniques like JSA, HAZOP or Risk Assessment to prevent accidents.

Methodologies of Process Hazard Analysis

Any process at the workplace can be successful only when the people working in that place have a common understanding of the risks accepted, if the understanding of the risks is not equal then meaningless discussions take place at that place and the result is that Could not get clarified.

It can be more effective only if following points are followed.

1. What if there are risks at work? This question should be considered from the start of the work.
2. Check list should be available for each job.
3. Work site hazard and work operability study should be done (HAZOP)
4. Fail Mode of Work and its Effects Analysis (FMEA)

Safety Measures in Design and Operation

4.1 Introduction

Safety measures can be adopted in both design and operational stages to avoid catastrophic incidents. One of the major events that can result in serious consequences in oil and gas industries is fire and explosion. With respect to control of accidents in process industry, the major focus is to prevent fire and explosion. While there are many methods to accomplish this, for a fire, accident, or combustion explosion to happen, three basic conditions must be fulfilled: (i) presence of combustive or explosive material; (ii) presence of oxygen to support the combustion reaction; and (iii) source to initiate the reaction. Absence of any one of the three may not induce a blast or burst. It is therefore imperative, though impractical, that one should design a process scheme by avoiding the presence of combustive material. This is due to the basic fact that offshore drilling operations deal with explosive material and a huge inventory of these in their storage compartments. Further, combustion mechanism can never be averted due to the presence of oxygen in abundance; this will rather support or initiate fire. By default, due to the electric processes present in an offshore and petroleum industry, source of kindling is an integral component of the organization. The fundamental question that we need to address is how can we design the system effectively?.

There are many preventive measures for fire and explosion. One of the common methods is inerting or purging. This is a mechanism by which the oxygen content or the fuel concentration is brought below a target value. For this purpose, flammability diagram is used to control the arms of the fire triangle. Static electricity present in the system can also be controlled using this method. Layout of the plant, with a good ventilation system can also help dilute the flammable mixture with more air concentration. Use of explosion-proof equipments and instruments are strongly recommended as one of the effective hazard control measures for fire and explosion. Well-designed sprinkler systems can also be one of the feasible solutions.

4.2 Inerting or Purging

Inerting or purging is a mechanism by which one can reduce the oxygen or fuel concentration below a specific target value. Usually, this is 4% below the limiting oxygen concentration (LOC). Nitrogen or carbon dioxide can also be used but nitrogen is commonly preferred in the purging process. Purging essentially reduces the oxygen concentration in the environment. Various methods are available for purging such as vacuum purging, pressure purging, combined purging, vacuum and pressure purging with impure nitrogen, sweep-through purging, and siphon purging.

4.4 Factors Affecting Purging

Replacement of one gas by another in an enclosed space or chamber takes place by means of two distinct actions: (i) displacement; and (ii) dilution or mixing. In a purge that is effected entirely by displacement; gas or air that is originally present in the container is pushed out of the escape vents by the entry of purge gas with little or no mixing of the purge gas with original contents. Thus, quantity of purge gas required for purging by displacement approximates the quantity of gas or air replaced. Certain conditions such as the size or shape of the chamber and the nature of gases cause the purge gas to mix with the original contents; this may lead to the dilution of purge gas. Purging by dilution can be accomplished in some situations by alternately pressurizing and depressurizing the facility. To accomplish a satisfactory purge by dilution or mixing, volume of inert purge gas that is required may be four or five times that of the free space of the chamber being purged. Increased demand in its contents is due to the fact that as purging proceeds, increasing amount of purge gas is lost from the escape vents in mixture with the original contents.

Almost all purging operations are combinations of displacement and dilution actions. In the actual practice, it is impossible to avoid mixing of the purge gas with air that is being replaced. In general, lesser the mixing or dilution more is the efficiency of purge. Purging, which proceeds with mixing or dilution that occurs in tanks and holders should be accomplished with an inert purge medium to avoid flammable mixtures. Purging without the use of an inert medium should be limited to pipelines where the amount of mixing can be controlled satisfactory by other methods.

4.6 Methods of Purging

There are many methods of purging that are commonly practiced in process industries. Broadly there are two types namely: batch purging and continuous purging. Some of the common methods of purging are as follows:

4.6.1 Siphon Purging

In this method, equipment that is to be purged is filled with liquid. Purged gas is introduced into the vapor space to replace the liquid, which is drained from the enclosure. Volume of the purge gas required in this method will be equal to the volume of the vessel and the rate of application will be the same as the rate of draining.

4.6.2 Vacuum Purging

In this method, equipment that normally operates at reduced pressure (or capable of operating at reduced pressure) is purged during the shutdown by breaking vacuum with the purge gas. If the initial pressure is not lower enough to ensure the desired low oxidant concentration, it becomes necessary to reevacuate and repeat the process. Amount of the purge gas required

in this case is determined by the number of applications required to develop the desired oxidant concentration. Where two or more containers or tanks are joined by a manifold and to be purged as a group, then the vapor content of each container or tank should be checked for completeness of purging.

4.6.3 Pressure Purging

In this method, enclosures might be purged by increasing the pressure within the enclosure. This is carried out by introducing the purge gas under pressure. After the gas has diffused, the enclosure is vented out to the atmosphere. In this case, more than one pressure cycle might be necessary to reduce the oxidant content to the desired percentage. Where two or more containers or tanks are joined by a manifold and to be purged as a group, then the vapor content of each container or tank should be checked to determine that the desired purging has been accomplished. Where a container filled with combustible material is to be emptied and then purged, purge gas is applied to the vapor space at a pressure consistent with equipment design limitations, thus accomplishing both the emptying of the vessel and the purging of the vapor space in the same process in parallel.

4.6.4 Sweep-Through Purging

This method involves introducing a purge gas into the equipment at one opening and letting the enclosure content escape to the atmosphere through another opening, thus sweeping out residual vapor. The quantity of purge gas required depends on the physical arrangement. Pipelines can be effectively purged with a little more volume of purge gas if the gas can be introduced at one end and the mixture can be released at the other in parallel. However, vessels require quantities of purge gas much in excess of their volume. If the system is complex, involving side branches through which circulation cannot be established, the sweep-through purging method might be impractical; in such cases pressure or vacuum purging is more appropriate. In this method, the total quantity required might be lesser than that for a series of steps of pressure purging. Further, four to five times of volumes of purge gas is required to almost completely displace the original mixture for ensuring a complete mixture.

4.6.5 Fixed-Rate Purging

This method involves a continuous introduction of purge gas into the enclosure at a constant rate. This should be sufficient to supply the peak requirements so that that complete protection is provided. The advantages of this

method are simplicity, lack of dependence on devices such as pressure regulators, and possible reduced maintenance. The main disadvantage is the continuous loss of product where the space contains a volatile liquid due to constant “sweeping” of the vapor space by the purge gas. Other demerits are increased total quantity of purge gas since it is supplied regardless of whether it is needed or not. Possible disposal problems that arise from the toxic and other effects are also few other undesirable consequences of this method.

4.6.6 Variable-Rate or Demand Purging

This method involves the introduction of purge gas into an enclosure at a variable rate that is dependent on demand. The advantage is that the purge gas is supplied only when actually needed. Therefore it is possible to completely prevent influx of air, if desired. The disadvantage is that the operation depends on the functioning of pressure control valves; however, it is difficult to maintain these valves as they sometimes operate at low pressure differentials.

3.9 Explosions

Explosion is a rapid release of energy causing development of pressure or shock waves. There are different types of industrial explosions as discussed next:

Confined Vapor Cloud Explosion (CVCE):

It is a type of explosion that happens in a vessel or in a confined space (e.g., inside a building). It is generally caused due to the release of high pressure or chemical energy.

Vapor Cloud Explosion (VCE):

It is a type of explosion caused by the instantaneous vapor cloud, which is formed in air due to the release of flammable chemicals into atmosphere.

Boiling Liquid Expanding Vapor Explosion (BLEVE):

BLEVE is caused due to the instantaneous release of a large amount of vapor through narrow opening under the pressurized condition (Tasneem Abbasi and Abbasi, 2007).

Vented Explosion (VE)

It is caused due to venting of chemicals at a high velocity.

Dust Explosion

Dust explosion is caused due to the rapid combustion of fine solid particles.

3.10 Explosion Characteristics

Explosion energy is dissipated in different forms: (i) pressure wave; (ii) projectiles; (iii) thermal radiation; and (iv) acoustic energy (Planas-Cuchi et al., 2004). Some of the types of energy, which is generated due to explosion are discussed next:

Blast wave is a shock wave in open air, which is generally followed by a strong wind. *Overpressure* of an object is a result of an impacting shock waves on any object. *Detonation* is a kind of explosion in which reaction front moves at a speed greater than that of sound in the given medium. *Deflagration* is a kind of explosion in which reaction front (energy front) moves at a lesser speed than that of sound in the given medium.

4.14 Fire Prevention

Safe storage of combustible and flammable materials and their appropriate location are important to ensure fire protection. Smoking is prohibited in the vicinity of the operations that constitute a pre-hazard. Such locations should be conspicuously posted with a sign board of *no smoking*. Changing rooms and other buildings where smoking is permitted should be located in areas designated safe for smoking. No source of ignition should be permitted in an area where smoking has been prohibited, unless it is first determined safe to do so by the supervisor in-charge or his designated representative. Potential sources of ignition should be permitted only in the designated areas located at a safe distance from the Well head or flammable liquid storage areas. Equipment, cellars, ground areas around and adjacent to the facility should be kept free from oil and gas accumulations that might create or aggravate pre-hazards. Combustible materials such as oily rags and waste should be stored in the covered metal containers. Natural gas or liquefied petroleum gas should not be used to operate spray guns or pneumatic tools.

4.15 Fire Protection

Fire-fighting equipments should not be removed for other than pre-protection, pre-fighting purposes and services. A pre-fighting water system should be used for wash down and other utility purposes so long as its pre-fighting capability is not compromised. Fire extinguishers and other pre-fighting equipments should be suitably located, readily accessible, and clearly labeled by their type and method of operation. Fire suppression equipments such as extinguishers, fixed systems etc. should be periodically inspected and maintained in operating condition at all times. A record of the most recent equipment inspection should be maintained. Portable pre-extinguishers shall be tagged with a durable tag showing the date of the last inspection, maintenance, recharge using the acceptable record keeping media. Inspection and maintenance procedures should comply with NFPA 10. Personnel should be

3.22 Sprinkler Systems

Sprinkler systems are also one of the effective means of controlling spread of fire. It is mandatory for all process plants to have a well-designed sprinkler system for ensuring safety to personnel and plants and equipments. A few sprinkler systems that are common in oil and gas industries are discussed in the next section:

3.22.1 Anti-freeze Sprinkler System

It consists of a wet-pipe system that contains an anti-freezing solution, which is connected to the water supply system. On demand, the sprinkler system opens the valves and pressurizes the liquid to control the fire hazard.

3.22.2 Deluge Sprinkler System

It consists of open sprinklers and an empty line that is connected to a water supply line through a valve. The valve is opened upon detection of heat and flammable material and water is sprayed through the nozzle of the sprinklers, which controls the spread of fire.

3.22.3 Dry Pipe Sprinkler System

This type of system is filled with nitrogen or air under pressure. When sprinkler nozzles are opened by heat, system is depressurized and allows water to flow into the system, which is further sprayed through the nozzles.

3.22.4 Wet Pipe Sprinkler System

This type of system contains water, which is discharged through nozzles that are opened when fire outbreaks in a plant. The design of sprinkler systems are based on the NFPA 69: Standards on Explosion prevention systems. As per NFPA norms, nominal discharge rate for 12.5 mm orifice spray nozzle are as shown in Table 3.5.

Table 3.5 Nominal discharge rate as per NFPA norms

Quantity of water (m ³ /s)	0.04	0.055	0.075	0.112	0.13
Pressure (N/mm ²)	0.069	0.138	0.24	0.52	0.69